0.1 Last Time

Let $D = \sum_\rho a_\rho D_\rho$ a T-invariant Cartier divisor on X_Σ. This comes from a line bundle $\mathcal{O}_X(D)$.

From every ray we can get a half-space and we can intersect all of them.

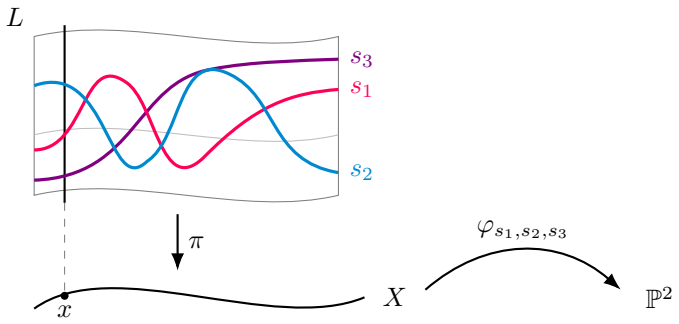
$$P_D = \bigcap_{\rho \in \Sigma(1)} \{m : \langle m, u_\rho \rangle \geq -a_\rho\} \subseteq M$$

There's a canonical bijection between lattice points of the polyhedron and T-equivariant regular sections of $\mathcal{O}_X(D)$. In particular

$$D_1 \sim D_2 \iff P_1 = P_2 + \widetilde{m}.$$

1 Maps to projective space

It's true that if have a space X, a line bundle $L \xrightarrow{\pi} X$, and a couple of (regular) sections s_1, s_2, s_3, the for every point in X, we have 3 points in the fiber. The ratio of any two of them is a well defined complex number. Another way to say this is to say that -that- triple is a set of homogeneous coordiantes in $\mathbb{P}^2$.



From here we have a map

$$\varphi_{s_1,s_2,s_3} : X \to \mathbb{P}^2, \quad x \mapsto [s_1(x) : s_2(x) : s_3(x)].$$

This is an abuse of notation because $s_i(x)$ is a vector, not a number. But it suffices to pick an isomorphism. This map is well defined when X is such that not all the sections we've chosen vanish at x. (This is basepoint freeness).

If we only want a map depending only on L, then we can

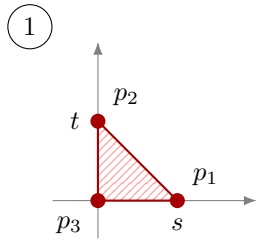
1. pick a basis $s_0, \ldots, s_r$ for $H^0(X, L)$ (space of regular sections of L and let $\varphi_L := \varphi_{s_0,\ldots,s_r}$.
2. $\varphi_L : X \to \mathbb{P}(H^0(X, L)^\vee)$ with $x \mapsto [\text{“ev}_X : H^0(X, L) \to k,\ s \mapsto s(x)\text{”}]$. This is highly undefined! So we should be screaming.

Definition 1. A line bundle L is basepoint free if for every $x \in X$, there exists $s \in H^0(X, L)$ such that $s(x) \neq 0$.

If $H^0(X, L)$ is finite dimensional, this implies that any basis $s_0, \ldots, s_r$ has no common zero. Hence it defines a morphism

$$\varphi_L : X \to \mathbb{P}^r, \qquad x \mapsto [s_0(x) : \cdots : s_r(x)].$$

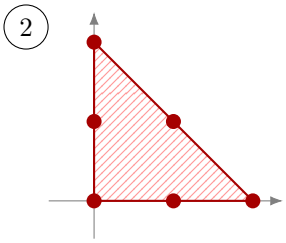
Example 1. (...) We have a map $\mathbb{P}^2 \to \mathbb{P}^2$ which restricted to the torus is the identity $(s,t,1) \mapsto (s,t,1)$.



$$\begin{array}{ccc} \mathbb{P}^2 & \xrightarrow{\text{id}} & \mathbb{P}^2 \\ \uparrow & & \uparrow \\ T & \xrightarrow{\text{id}} & T \end{array}$$

$$[s:t:1] \longmapsto [s:t:1]$$

Example 2. We could remain in $\mathbb{P}^2$ and scale the triangle by 2!



$$\begin{array}{ccc} \mathbb{P}^2 & \xrightarrow{\nu_2} & \mathbb{P}^5 \\ \uparrow & & \uparrow \\ T & \longrightarrow & T_5 \end{array}$$

$$[s:t:1] \longmapsto [1:s:t:s^2:st:t^2]$$

Here T_5 denotes the dense torus of $\mathbb{P}^5$. This is the polytope of the divisor $2D$, where $D = \{z = 0\}$. We get the map

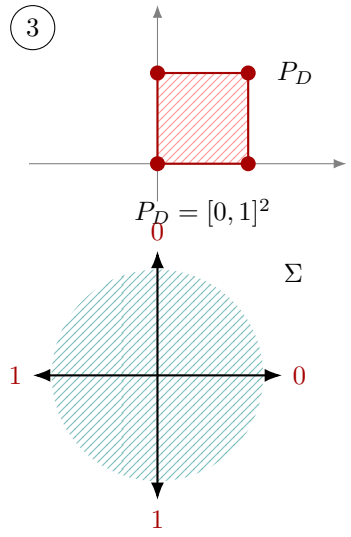
$$\nu_2 : \mathbb{P}^2 \to \mathbb{P}^5, \quad T \to T_5, \quad [s:t:1] \mapsto [1:s:t:s^2:st:t^2].$$

If we think of $(s,t,1) \cdot (x,y,z) = (sx,ty,z)$ then the map at the level of projective spaces is

$$[x:y:z] \mapsto [z^2:zx:zy:x^2:xy:y^2].$$

Here $\varphi(\mathbb{P}^2)$ is the Veronese surface in $\mathbb{P}^5$.

Example 3. Consider now this polytope.



- From the right ray we get $x \geq 0$,
- The top ray, $y \geq 0$.
- Left gives $-x \geq -1$ or $x \leq 1$, and
- Down gives $-y \geq -1$ or $y \leq 1$.

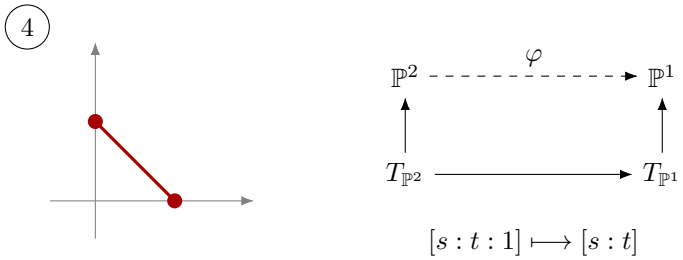
This is $X_\Sigma = \mathbb{P}^1 \times \mathbb{P}^1$. The map is

$$\begin{aligned} \varphi : \mathbb{P}^1 \times \mathbb{P}^1 &\longrightarrow \mathbb{P}^3, \\ T_{\mathbb{P}^1 \times \mathbb{P}^1} &\longrightarrow T_{\mathbb{P}^3}, \\ ([s:1],[t:1]) &\longmapsto [1:s:t:st]. \end{aligned}$$

At the level of projective spaces this map is

$$((x_1,x_2),(y_1,y_2)) \mapsto (x_2y_2, x_1y_2, x_2y_1, x_1y_1).$$

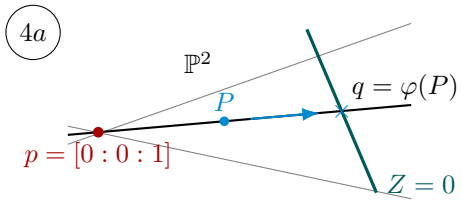
Example 4. Let's start breaking things.



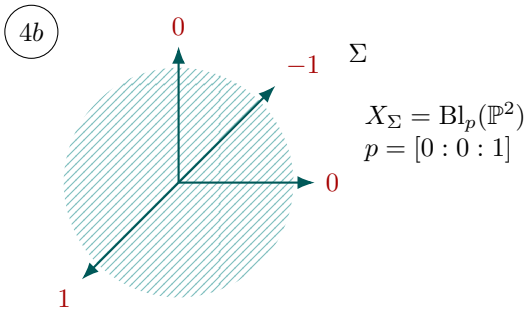
Consider the map

$$\varphi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^1,$$
$$T_{\mathbb{P}^2} \longrightarrow T_{\mathbb{P}^1}, \qquad [s:t:1] \longmapsto [s:t].$$

This corresponds to $[X:Y:Z] \mapsto [X:Y]$ which is not defined at $[0:0:1]$. This is a *rational* map, not regular. In classical geometry, this is projection from $p = [0:0:1]$ onto the line $Z = 0$. Every other point P determines the unique line pP, which meets $Z = 0$ at its image. This works everywhere except at p.



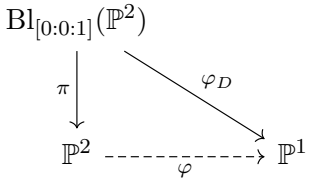
One could complain and say diagram 4 doesn't come from a divisor, but instead we say *picky picky* and consider diagram 4b.



This is the blowup of $\mathbb{P}^2$ at a point, $X_\Sigma = \mathrm{Bl}_{[0:0:1]}(\mathbb{P}^2)$.

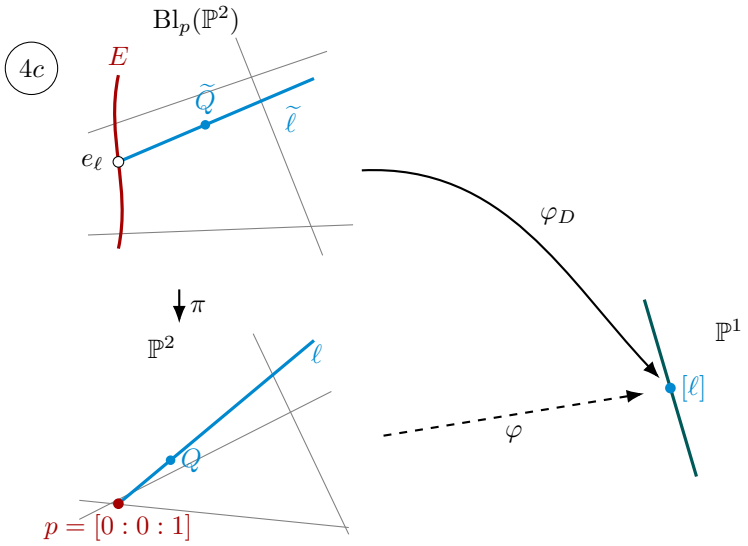
- The right ray gives $x \geq 0$,
- the $(1,1)$ ray gives $x + y \geq 1$.
- The top one gives $y \geq 0$ and
- $(-1,-1)$ gives $x + y \leq 1$.

And so here we get the same kind of polytope. The maps are



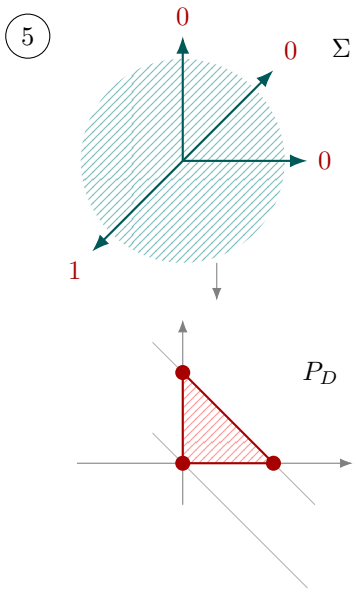
$$[X : Y : Z] \longmapsto [X : Y].$$

The lift φ_D is a regular morphism, but it is not injective.



Away from the exceptional divisor $E = \pi^{-1}(p)$, the lift agrees with the projection $[X : Y : Z] \mapsto [X : Y]$. A point $e_\ell \in E \cong \mathbb{P}^1$ records a tangent direction at p, equivalently a line ℓ through p. The map φ_D sends it to the corresponding point $[\ell]$ of the target, so $\varphi_D|_E : E \to \mathbb{P}^1$ is an isomorphism. The blowup map π contracts all of E to p. The map φ_D is not injective because each fiber is the strict transform $\widetilde{\ell}$ of a line through p, a copy of $\mathbb{P}^1$.

Example 5. Keep the blowup fan and assign coefficient 1 to $(-1,-1)$ and 0 to the other rays. The polytope is the standard triangle.



The divisor is the pullback of a hyperplane, and the associated map is the blowdown

$$\mathrm{Bl}_{[0:0:1]}(\mathbb{P}^2) \xrightarrow{\pi} \mathbb{P}^2.$$

This is a regular birational morphism: it restricts to an isomorphism away from E and $p = [0:0:1]$. It is not an isomorphism because it contracts the entire exceptional curve $E \cong \mathbb{P}^1$ to p. This divisor is basepoint free but not ample, since the maps associated to all positive multiples still contract E.

We can summarize the relation between divisors and their associated maps as follows. Let D be a Cartier divisor on a projective variety X, and use the complete linear system $|D|$.

- **Rational map.** When $|D|$ is nonempty, it determines a rational map φ_D. Its section formula is defined wherever the global sections of $\mathcal{O}_X(D)$ do not all vanish.
- **Basepoint free.** The divisor D is basepoint free if this section formula is defined at every point. It then gives a regular morphism on all of X.
- **Ample.** The divisor D is ample if some positive multiple mD is very ample, equivalently if φ_{mD} is a closed embedding for some integer $m > 0$.
- **Very ample.** The divisor D is very ample if φ_D itself is a closed embedding, identifying X with its closed image in projective space.